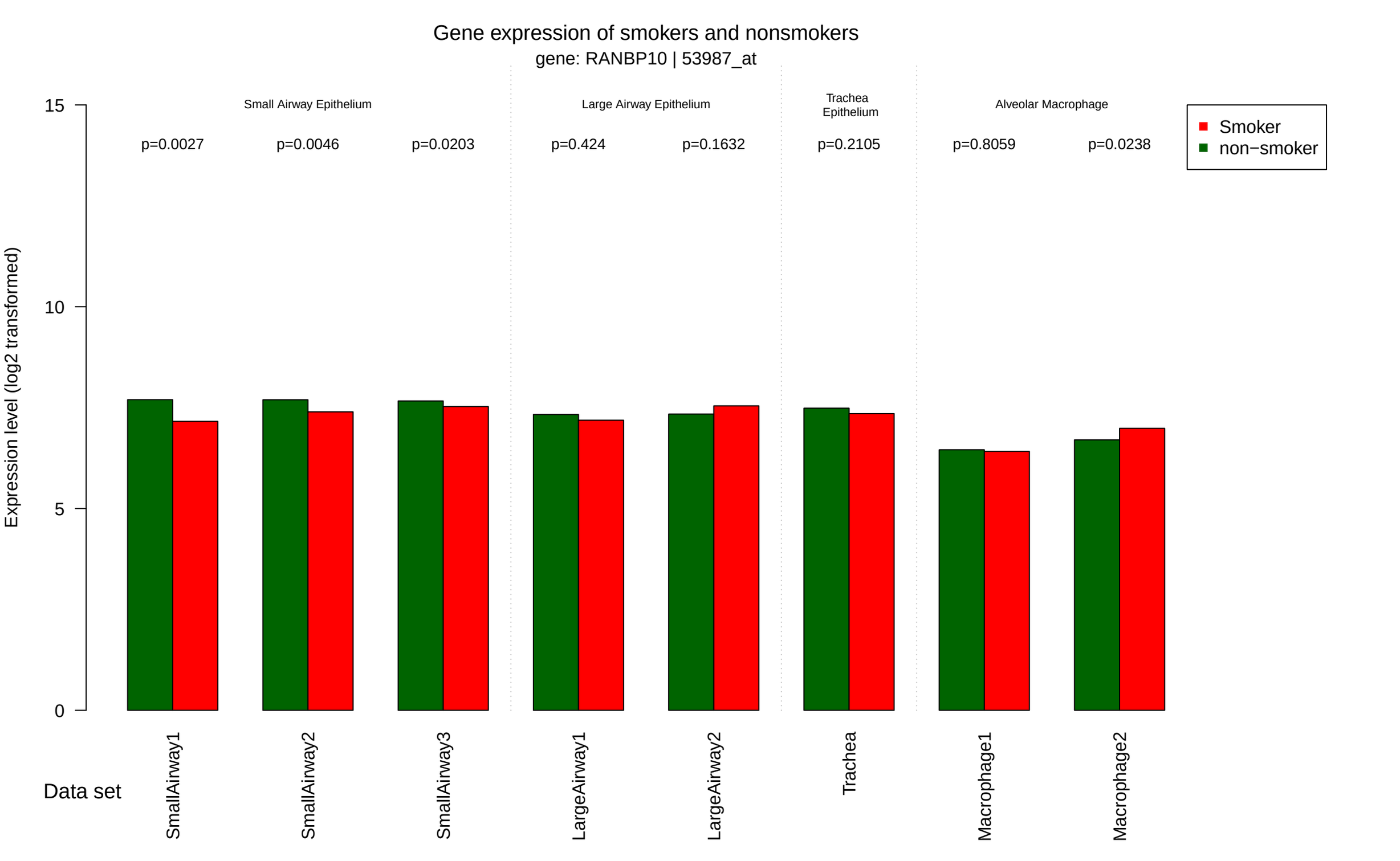
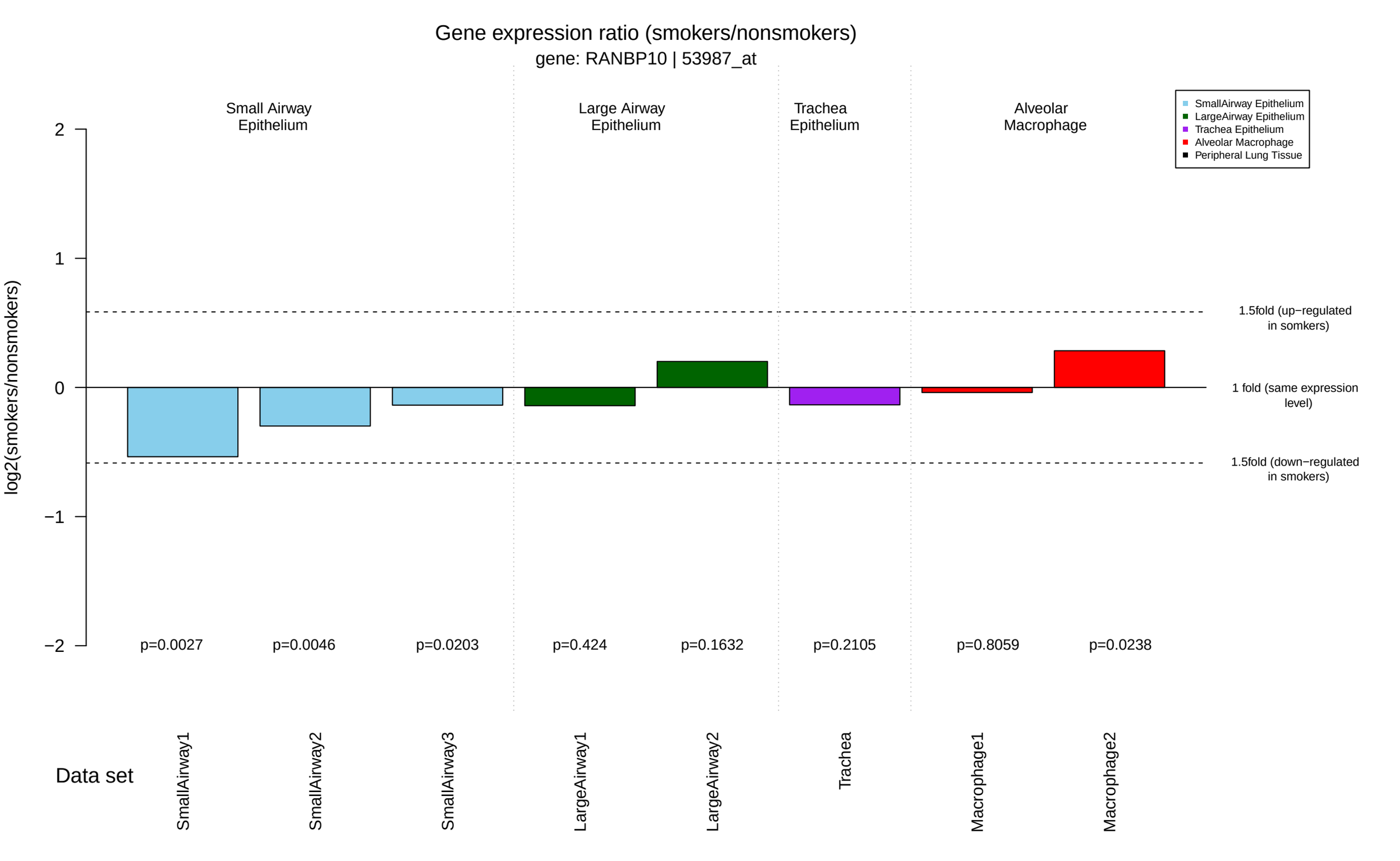
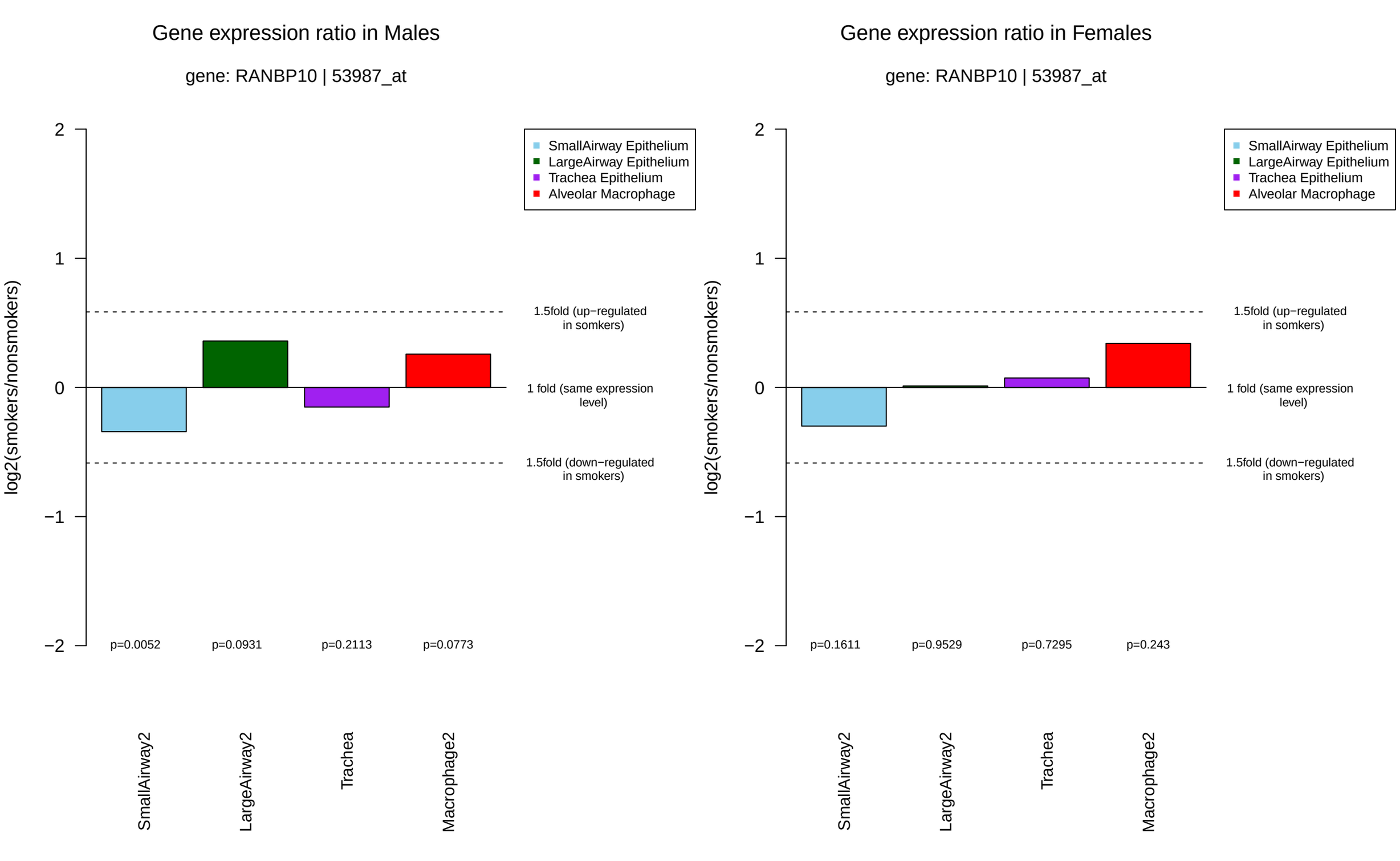
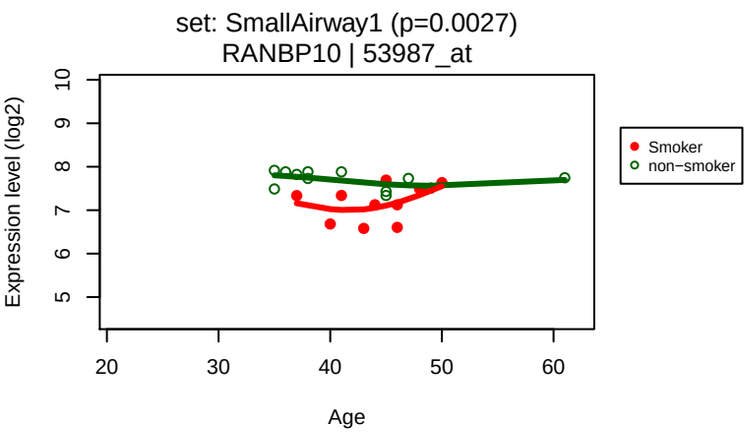
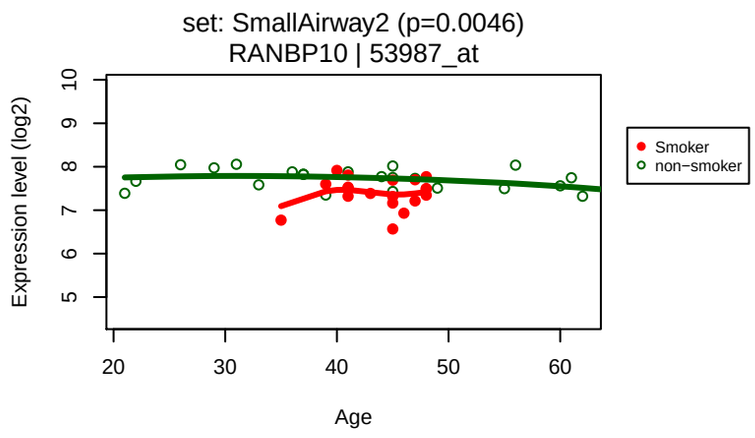
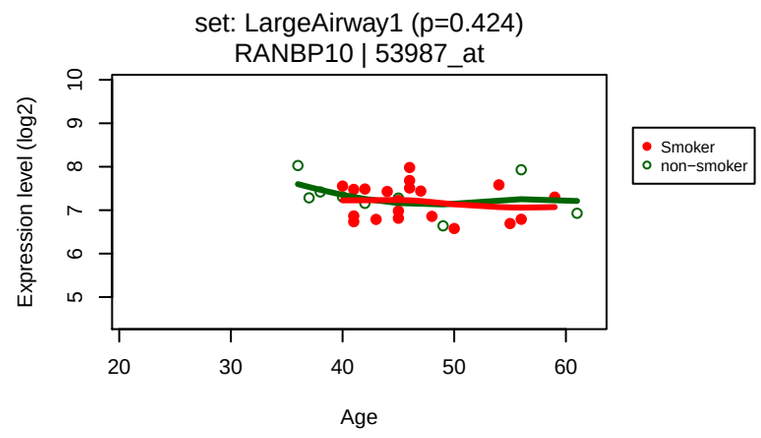

Gene expression of smokers and nonsmokers
gene: RANBP10 | 53987_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0027
p=0.0046
p=0.0203
p=0.424
p=0.1632
p=0.2105
p=0.8059
p=0.0238
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

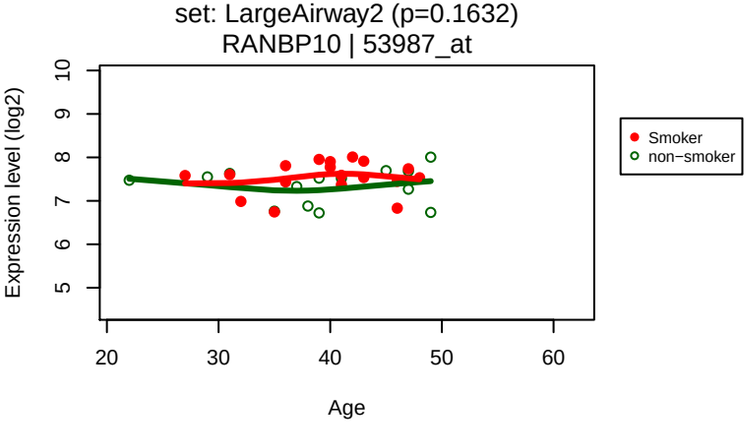

Gene expression ratio (smokers/nonsmokers)
gene: RANBP10 | 53987_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0027
p=0.0046
p=0.0203
p=0.424
p=0.1632
p=0.2105
p=0.8059
p=0.0238
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

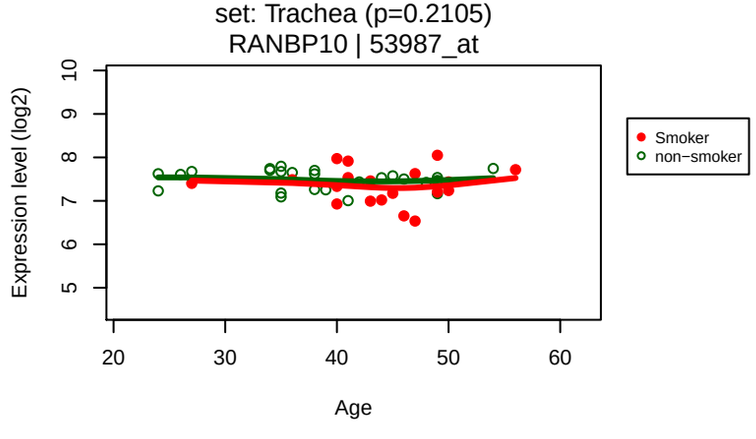

Gene expression ratio in Males
gene: RANBP10 | 53987_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0052
p=0.0931
p=0.2113
p=0.0773
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RANBP10 | 53987_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1611
p=0.9529
p=0.7295
p=0.243
SmallAirway2
LargeAirway2
Trachea
Macrophage2

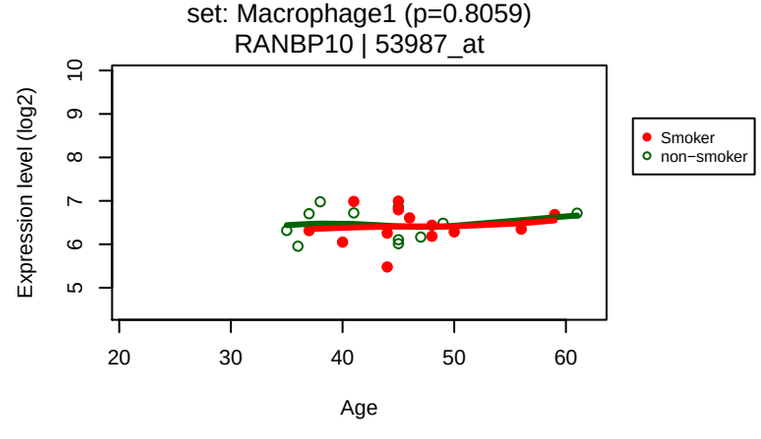

set: SmallAirway1 (p=0.0027)
RANBP10 | 53987_at
Expression level (log2)
Age
Smoker
non−smoker

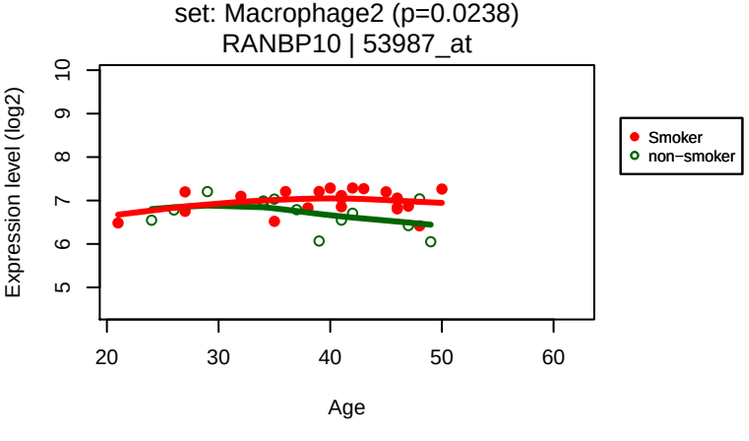

set: SmallAirway2 (p=0.0046)
RANBP10 | 53987_at
Expression level (log2)
Age
Smoker
non−smoker

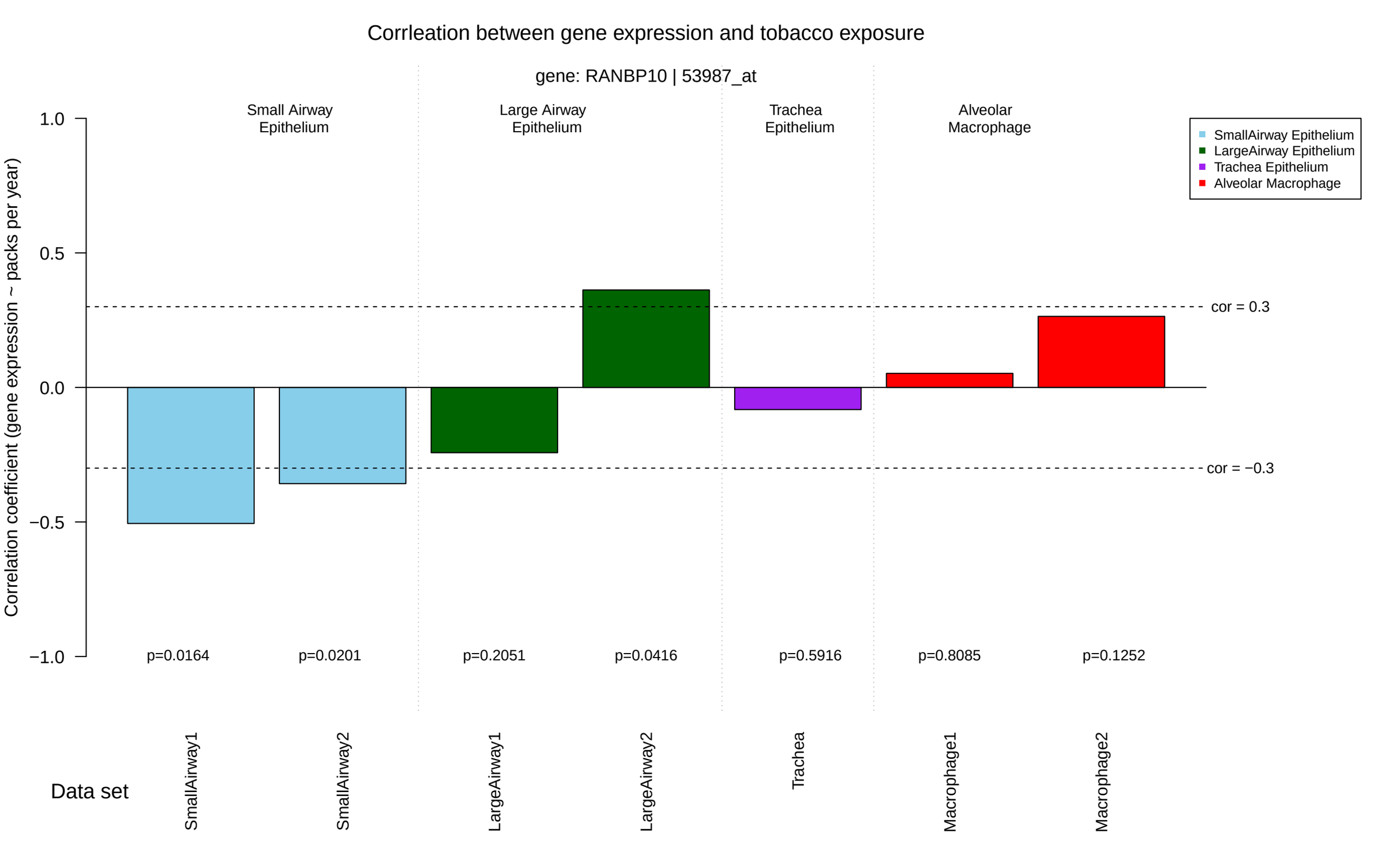

set: LargeAirway1 (p=0.424)
RANBP10 | 53987_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1632)
RANBP10 | 53987_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2105)
RANBP10 | 53987_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8059)
RANBP10 | 53987_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0238)
RANBP10 | 53987_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RANBP10 | 53987_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0164
p=0.0201
p=0.2051
p=0.0416
p=0.5916
p=0.8085
p=0.1252
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2